[NEW RESULTS]-KDD-1060

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1 Additional Experiment Results to Reviewer gq6s

2. Interpretability Validation (W1, W5)

We evaluate real-data interpretability using the `I(20)` and `I(100)` metrics. Concretely, all explainers are evaluated under the same ranking, masking, and retraining pipeline with a shared evaluator: `I(20)` reports predictive performance after retaining the top-ranked 20% evidence budget, whereas `I(100)` reports the corresponding full-budget score. Higher scores combined with lower average ranks indicate that the highlighted temporal evidence effectively preserves predictive signals.

Table 1 summarizes these faithfulness results under a fixed masking and retraining protocol. Rather than claiming complete semantic interpretation, our objective is to verify whether top-ranked temporal regions retain actual predictive power across complex domains. Under the stringent `I(20)` threshold, AnchorMoE achieves 0.6419, exceeding both the strongest time-series specific baseline (FIT, 0.5760) and the generic baseline (IG, 0.6189). Under `I(100)`, AnchorMoE reaches 0.8620, and its gradient-free variant, AnchorMoE (w/o grad), reaches 0.8667, both outperforming Dynamask (0.8284). These results confirm that the temporal evidence identified by our model remains highly predictive on real data.

Figure 1 complements these quantitative results by providing a representative NATOPS case study. It demonstrates that the empirical faithfulness gains are naturally accompanied by a compact and localized temporal focus, effectively avoiding diffuse attribution across the sequence.

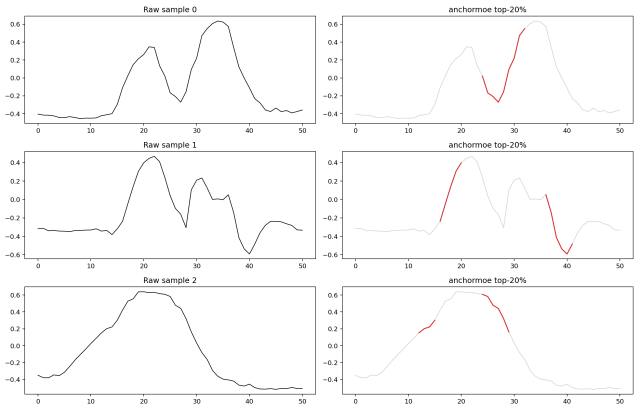

Figure 1: Representative NATOPS case study. Temporal evidence highlighted by AnchorMoE remains localized and compact.

2 Additional Experiment Results to Reviewer Hjgq

1. Principal Explanation (W1, W3, W4, W5)

To address concerns regarding the function of gradients within our framework, we isolate their role during training. Referring back to the global evaluation in Table 1, AnchorMoE (w/o grad) still comprehensively exceeds the Integrated Gradients baseline on the mean `I(100)` score (0.8667 vs. 0.8384). However, the standard gradient-calibrated AnchorMoE model maintains an advantage under the more restrictive `I(20)` threshold. This performance pattern aligns directly with our design motivation: gradient magnitude acts as an effective instance-level calibration signal during the training phase, rather than serving as the final extracted explanation at inference.

2. Internal Mechanism (W2, W7, W8, W9)

To address concerns regarding potential expert collapse, we comprehensively tracked the internal utilization statistics across our models. The active expert rate defines the fraction of available experts utilized, while effective experts measure the diversity-adjusted utilization. In this context, representation collapse would manifest as a critically low active rate, near-unit max share, and effective experts approaching 1.0, whereas healthy specialization preserves non-trivial utilization without enforcing a uniform regime.

Table 2 reports the aggregate statistics across all 29 datasets. Expanding on this, the visual distributions in Figure 2 demonstrate that the routing concentration and confidence sparsity dynamically adapt to dataset characteristics rather than collapsing into a single pattern.

To consolidate the structural necessity of our design, Table 3 provides a comprehensive view of our targeted internal-design ablations: B1 probes the MVSE query components, B2 probes the DPAR routing prior, and B3 probes UADB confidence-weighted binding.

For the MVSE query formulation (B1), removing the sample-level relevance cue causes the mean accuracy to degrade significantly from 0.9686 to 0.7084, whereas ablating the spectral cue maintains the block average. This behavior accurately reflects our architectural intent: the relevance cue is primarily responsible for injecting sequence-level context into the routing decisions. For the structural necessity of our routing mechanism (B2), we observe that removing DPAR increases the inter-anchor redundancy from 0.3488 to 0.7213. This 106.8% surge confirms that DPAR functions primarily as a redundancy-reducing geometric prior that enforces diverse specialization, rather than acting as a simplistic traffic-balancing mechanism.

3 Additional Experiment Results to Reviewer gMhY

1. Scalability Analysis & Runtime (W1)

To comprehensively address scalability concerns, we re-evaluate the framework focusing on the end-to-end throughput required for transparent decision-making. We rerun the benchmark under a unified synthetic 6-class multivariate-classification protocol: all deep models are trained for 1 FP32 epoch on a single GPU with batch size 4. In each sweep, one axis among `feature_dim`, `seq_len`, and `num_samples` varies over {10, 100, 1000, 10k, 20k, 40k}, while the

Table 1: Real-data faithfulness evaluation. Higher is better for `I(20)` and `I(100)`; lower is better for rank.

Dataset / Stat.	Metric	AnchorMoE	AnchorMoE (w/o grad)	IG	DeepLift	Grad-CAM	Random	Dynamask	FIT
Overall Mean	I(20)	**0.6419**	**0.6338**	0.6189	0.6168	0.5891	0.5301	0.5390	0.5760
	I(100)	**0.8620**	**0.8667**	0.8384	0.8363	0.8278	0.8035	0.8284	0.8264
Overall Rank	I(20)	**2.86**	3.43	**3.14**	3.86	4.71	6.29	6.00	5.71
	I(100)	**2.57**	**2.86**	3.71	4.43	5.86	6.29	4.57	5.71
ArticularyWordRec.	I(20)	0.6554	0.6625	**0.6808**	**0.6749**	0.5832	**0.6749**	0.5757	0.6361
	I(100)	0.8772	0.8810	0.8812	**0.8963**	0.8599	**0.8938**	0.8812	0.8725
BasicMotions	I(20)	**0.5750**	**0.5938**	0.4625	0.5094	0.5521	0.4583	0.5083	0.4531
	I(100)	**0.8175**	**0.8675**	0.7354	0.7306	0.7792	0.7388	0.8092	0.7819
Cricket	I(20)	0.6597	0.6111	**0.6985**	**0.6997**	0.6140	0.6198	0.6024	0.6869
	I(100)	0.8993	0.8819	**0.9041**	**0.9078**	0.8703	0.8443	0.8760	0.8892
FordA	I(20)	0.6661	**0.6821**	0.6301	0.6412	0.6469	0.4620	**0.6920**	0.5631
	I(100)	0.8678	**0.8766**	0.8401	0.8287	0.8541	0.7505	**0.8800**	0.8061
NATOPS	I(20)	0.6764	0.6708	**0.6792**	**0.6769**	0.6069	0.5315	0.4843	0.5486
	I(100)	**0.8883**	**0.8808**	0.8744	0.8671	0.8517	0.8500	0.7591	0.8252
RacketSports	I(20)	**0.5576**	0.5206	0.4729	0.4482	**0.5414**	0.4238	0.4833	0.4512
	I(100)	**0.7566**	**0.7541**	0.7054	0.7053	0.7194	0.6762	0.7278	0.6910
SpokenArabicDigits	I(20)	**0.7030**	0.6959	**0.7081**	0.6675	0.5790	0.5405	0.4271	0.6928
	I(100)	**0.9274**	0.9251	**0.9283**	0.9180	0.8600	0.8711	0.8658	0.9191

Mechanism Diagnostics from the 29-Dataset Sweep

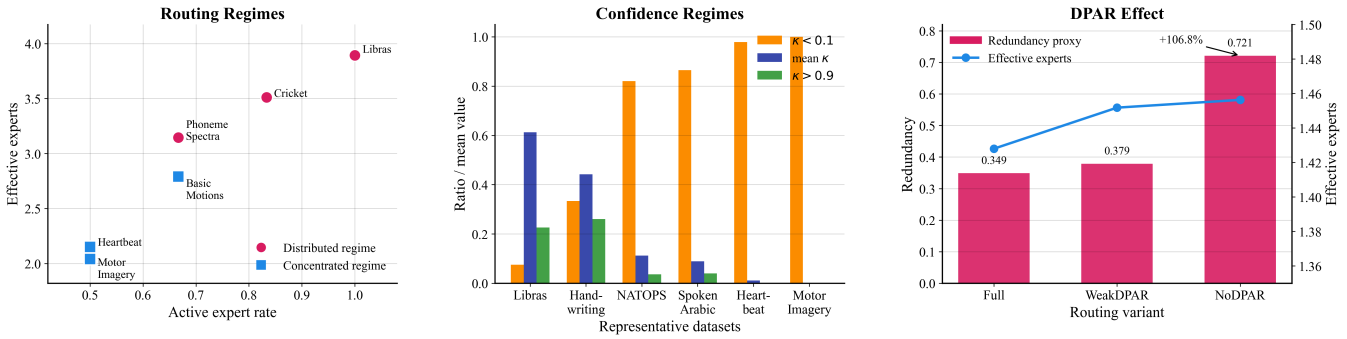

Figure 2: Mechanism diagnostics across 29 datasets, highlighting dataset-dependent utilization and confidence behavior.

Table 2: Aggregate mechanism statistics across the 29-dataset diagnostic sweep. Effective expert counts consistently remain above 2.0, confirming the absence of severe representation collapse.

Metric	Mean	Median	Min	Max
Active Rate	0.6388	0.6000	0.3333	1.0000
Max Share	0.3674	0.3578	0.2705	0.4950
Effective Experts	2.8952	2.9053	2.0412	3.8933
Mean κ	0.1844	0.1662	0.0001	0.6128
$\kappa < 0.1$ (Low)	0.5653	0.6325	0.0000	1.0000
$\kappa > 0.9$ (High)	0.0272	0.0002	0.0000	0.2603

other two are fixed to 100. AnchorMoE extracts its intrinsic explanation seamlessly during the routed forward pass, whereas every baseline (including MiniRocket) is charged with a fixed-budget temporal occlusion procedure using 10 windows. Each run is capped at 400 seconds. We record explicit timeout (TO) and out-of-memory (OOM) occurrences across these three axes. Raw peak-memory logs are retained in the released benchmark files, while Table 4 and Figure 3 focus on direct throughput and task completion behavior.

Table 4 and Figure 3 detail the results of this extended benchmark. Because AnchorMoE bypasses the computational tax associated with post-hoc attribution, it successfully outperforms the strongest completed baseline at each point in **16 of 18** evaluated configurations. It demonstrates particular dominance along the sequence length and sample size axes, winning all test points. For instance, at the extreme 40,000 sequence length mark, AnchorMoE achieves a 1.48× speedup, processing 4.80 iterations per second compared to 3.25 for TSLANet. Similarly, at 40,000 samples, it processes 227.13 iterations per second against 188.35 for DLinear. Furthermore, as depicted in the corresponding figures, certain baselines eventually trigger timeout or out-of-memory errors at these extreme scales, whereas our framework remains stable. We acknowledge that DLinear remains exceptionally competitive purely along the feature dimension sweep due to its minimal linear operator. However, once explanation costs are fairly accounted for, our framework maintains a clear and consistent efficiency advantage on the temporal and volume dimensions that most directly stress real-world deployments.

Table 3: Comprehensive internal-design ablation and mechanism diagnostics. Experiments are grouped into discrete blocks (B1: MVSE, B2: DPAR, B3: UADB). Metrics marked with '–' are invariant or non-applicable to the specific diagnostic block. Because diagnostic subsets vary across blocks, absolute scores differ and are not directly comparable to the 18-dataset overall averages in the main paper.

Block	Variant	Mean Acc	I(20)	I(100)	Eff. Experts	Redundancy	Mean κ
B1	**Full**	0.9686	0.9250	0.9348	–	–	–
	NoRelevance	0.7084	0.8478	0.9348	–	–	–
	NoSpectral	0.9737	0.9268	0.9348	–	–	–
B2	**Full**	0.9686	0.9250	0.9348	2.4280	0.3488	0.5319
	NoDPAR	0.9727	0.9580	0.9348	2.4563	0.7213	0.5600
	WeakDPAR	0.9734	0.9518	0.9348	2.4518	0.3788	0.5306
B3	**Full**	0.9509	0.8953	0.9081	–	–	0.3833
	AnchorMoE (w/o grad)	0.9638	0.9097	0.9051	–	–	0.2322
	NoConfidence	0.9477	0.8819	0.9078	–	–	1.0000

Table 4: Throughput comparison (iter/s) under the unified protocol. The best and second-best throughputs in each row are highlighted. 'TO' and 'OOM' denote timeout and out-of-memory failures. AnchorMoE visually dominates the throughput across extreme temporal and sample scales.

Axis	Scale	AnchorMoE	PatchTST	FED former	DLinear	LightTS	MICN	TimesNet	TSLANet	MPTSNet	Mini Rocket
					Feature-dimension sweep						
Feature Dim.	10	**273.303**	36.039	0.920	**93.836**	45.451	26.393	12.059	11.181	0.278	0.120
	100	**218.865**	49.023	2.709	**144.414**	45.171	32.882	13.089	45.296	0.261	0.188
	1000	**111.366**	14.323	2.697	**83.065**	37.089	30.262	14.627	39.410	0.243	0.072
	10k	**15.882**	0.982	0.693	**16.532**	6.510	12.483	3.968	5.977	0.081	0.056
	20k	**8.879**	TO	1.940	**8.324**	0.892	6.232	4.528	6.817	0.268	0.156
	40k	3.374	OOM	1.636	**3.898**	OOM	3.459	2.778	**3.853**	0.252	0.148
					Sequence-length sweep						
Sequence Len.	10	**237.871**	42.578	12.226	**107.408**	85.805	37.816	12.334	47.824	1.672	0.472
	100	**215.164**	44.297	2.737	**185.154**	56.326	30.299	16.623	46.015	0.257	0.188
	1000	**94.381**	9.466	2.683	**73.020**	44.992	27.633	5.985	33.941	0.029	0.167
	10k	**16.198**	OOM	TO	4.275	4.655	TO	TO	**12.136**	OOM	0.156
	20k	**7.347**	OOM	TO	OOM	1.516	TO	TO	**6.896**	OOM	0.148
	40k	**4.804**	OOM	TO	OOM	OOM	TO	TO	**3.249**	OOM	0.130
					Number-of-samples sweep						
Num. Samples	10	**107.708**	20.893	2.390	**72.935**	51.193	24.553	8.103	32.451	0.283	0.398
	100	**234.845**	44.372	2.702	**162.943**	60.522	36.469	16.167	47.770	0.239	0.187
	1000	**265.869**	57.499	2.539	**150.518**	62.998	37.092	14.474	48.258	0.275	0.189
	10k	**250.911**	52.411	TO	**192.657**	77.862	33.067	15.747	44.341	TO	TO
	20k	**262.604**	53.769	TO	**171.634**	69.332	36.402	15.688	46.513	TO	TO
	40k	**227.131**	52.324	TO	**188.349**	72.191	34.161	TO	40.647	TO	TO

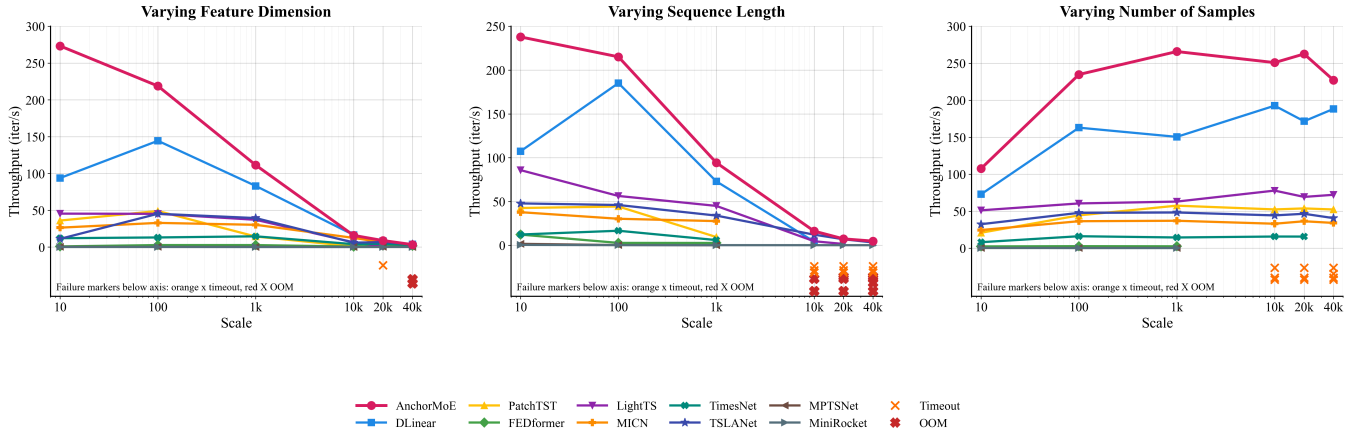

Figure 3: Explanation-aware throughput trends under the unified protocol. Solid lines denote completed runs, while markers below the axis indicate timeout or out-of-memory failures.

2. Interpretability on Real Data (W2)

As evaluated in Section 1, we employ the strict `I(20)` and `I(100)` metrics to measure predictive temporal localization. For the exact evaluation rule, we follow the identical ranking, masking, and retraining pipeline defined in Section 1. The corresponding results in Table 1 and Figure 1 confirm that our framework consistently

preserves structural integrity and maintains a compact temporal focus on real-world datasets.

4 Additional Experiment Results to Reviewer ncgk

1. Tuning Complexity & Anchor Determination (W1, W3)

To assess tuning sensitivity, Table 5 records the model response across anchor diversity, spectral weight, group count, and confidence regularization. The empirical evidence counters the assumption of uniform brittleness. While structural factors such as anchor diversity and spectral weight present relatively stable operating regions, the confidence regularization weight requires conservative tuning. Specifically, on the more difficult NATOPS dataset, aggressive confidence regularization drives accuracy sharply from **0.9500** to **0.1667**, as reflected by the `conf_lambda` sensitivity evidence summarized in Table 5 and Figure 4.

To mitigate this, we propose a reproducible, saturation-driven heuristic for determining the optimal number of groups: selecting the minimal count that achieves validation plateau. As supported by Table 6, this dataset-specific optimum consistently falls within a moderate and predictable range of 3 to 5 groups.

Table 5: Sensitivity analysis summary detailing performance boundaries under key structural hyperparameters.

Parameter	Scanned values	Acc. Range
Anchor Div.	0, 0.02, 0.05, 0.08, 0.1	0.9268–0.9332
Spectral Wt.	0, 0.1, 0.3, 0.5, 0.7	0.9281–0.9374
Num. Groups	2, 3, 4, 5, 6, 7, 8	0.9166–0.9722
Conf. λ	0, 0.005, 0.01, 0.02, 0.03	0.7976–0.9314

Table 6: Dataset-specific optimal group counts, confirming that performance consistently saturates within a moderate range.

Dataset	Best num_groups	Acc	F1	AOPCAcc	IoU
ArticularyWord Recognition	3	0.9733	0.9733	0.7898	0.2820
Cricket	5	0.9722	0.9720	0.7000	0.2796
FordA	3	0.9356	0.9356	0.3461	0.2499
Heartbeat	4	0.7707	0.7674	0.4898	0.2769
NATOPS	4	0.9833	0.9833	0.6578	0.3750
SpokenArabic Digits	3	0.9900	0.9900	0.7773	0.3007

2. Risk of Over-filtering & Safety-Critical Trade-offs (W2)

Building on the real-data evaluations, the B3 block of Table 3 investigates the risk of over-filtering by ablating the intrinsic confidence score entirely. Without this self-assessment capability, the model reverts to a uniform gating regime ($\kappa = 1.0$), treating every routed patch identically regardless of its actual informativeness. As background noise dilutes the genuine discriminative signals, the strict early-stage feature localization score, `I(20)`, naturally drops from 0.8953 to 0.8819. This structural degradation validates our core design premise: while the routing mechanism governs spatial feature allocation, the confidence score provides an indispensable filtering layer to suppress weak evidence. These diagnostics jointly map out a safe and effective operating boundary for deploying the framework.

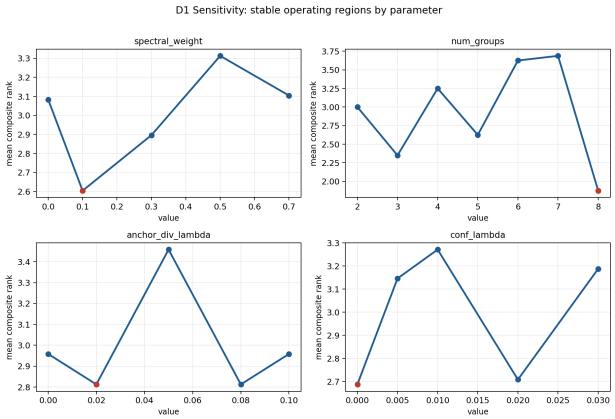

Figure 4: Stable operating regions for primary structural hyperparameters, with `conf_lambda` exhibiting the sharpest degradation boundary.